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Electronic Devices and Circuits Lab Final Project - MicroPhone Amplifier

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1 Objective

The objective of this project is to design, implement, and evaluate an audio amplification system consisting of a voltage amplifier stage followed by a Class-AB power amplifier. The system amplifies low-level audio signals from an auxiliary input to a level sufficient to drive a small loudspeaker while maintaining low distortion, stable biasing, and acceptable efficiency. Key performance parameters such as gain, bandwidth, output power, quiescent current, efficiency, noise, and distortion are analyzed through theoretical calculations and experimental measurements to validate the suitability of the Class-AB configuration for audio applications.

2 Theory

2.1 Overall Amplifier Architecture

- The amplifier system consists of two cascaded sections:
 - Dual-stage common-emitter voltage amplifier (Q1 and Q2)
 - Class-AB power amplifier stage
- The voltage amplifier provides high voltage gain.
- The power amplifier provides high current gain.
- Cascading ensures proper gain distribution and load isolation.

2.2 Dual-Stage Common-Emitter Pre-Amplifier

- The pre-amplifier uses two BC547B NPN transistors:
 - First CE stage: Q1
 - Second CE stage: Q2
- Both stages operate in active region.
- Each stage provides moderate voltage gain.
- Overall gain is obtained by cascading the stages.

Small-Signal Voltage Gain of First Stage

$$g_{m1} = \frac{I_{C1}}{V_T} \quad (1)$$

$$A_{v1} = -g_{m1}R_3 \quad (2)$$

- Negative sign indicates 180° phase inversion.

Second CE Stage (Q2)

- Input is coupled from Q1 collector using capacitor C1.
- Biasing resistors: R5 and R7
- Collector resistor: R8
- Emitter resistor: R6

$$A_{v2} = -g_{m2}R8 \quad (3)$$

Overall Pre-Amplifier Gain

$$A_{v(pre)} = A_{v1} \times A_{v2} \quad (4)$$

- Cascading improves gain without instability.
- Gain is sufficient to drive the power amplifier stage.

Role of Coupling Capacitors

- C2: Couples input signal to Q1 base.
- C1: Couples Q1 collector to Q2 base.
- Output capacitor: Couples signal to next stage.

$$f_L = \frac{1}{2\pi R_{eq}C} \quad (5)$$

Need for a Power Amplifier

- CE stages provide high voltage gain but low current capability.
- Load impedance is low compared to collector resistance.
- Direct driving causes distortion and signal loss.
- A power amplifier is required for current amplification.

2.3 Class-AB Power Amplifier

- Output stage operates in Class-AB mode.
- Uses complementary push-pull configuration.
- Output transistors conduct slightly more than 180° .
- Eliminates crossover distortion present in Class-B.

Quiescent Current

$$I_Q = \frac{V_{bias} - 2V_{BE}}{R_E} \quad (6)$$

- Prevents crossover distortion.
- Must be optimized to reduce power dissipation.

Output Characteristics

$$A_{v(power)} \approx 1 \quad (7)$$

$$R_{out} \approx \frac{R_E}{\beta} \quad (8)$$

- Voltage gain is unity.
- Current gain is high.

Efficiency of Class-AB Amplifier

$$\eta = \frac{P_{out}}{P_{DC}} \times 100\% \quad (9)$$

- Higher than Class-A.
- Lower than ideal Class-B.

Justification for Using Class-AB

Class	Distortion	Efficiency	Suitability
A	Very Low	Low	Poor
B	High	High	Poor
AB	Low	Moderate	Excellent

Table 1: Comparison of amplifier classes

2.4 Biasing Theory of the Dual-Stage CE Pre-Amplifier

Purpose of Biasing

- Biasing establishes a stable DC operating point (Q-point).
- Ensures transistor operation in the active region.
- Prevents cutoff and saturation during signal variations.
- Minimizes distortion and improves linearity.

Voltage Divider Biasing of First Stage (Q1)

- Biasing resistors $R1$ and $R2$ form a voltage divider.
- Divider provides a fixed base voltage independent of β .
- Collector resistor $R3$ sets collector voltage.
- Emitter resistor $R4$ provides negative feedback.

Base Voltage

$$V_{B1} = V_{CC} \cdot \frac{R2}{R1 + R2} \quad (10)$$

Emitter Voltage

$$V_{E1} = V_{B1} - V_{BE} \quad (11)$$

- $V_{BE} \approx 0.7 V$ for silicon transistor.

Emitter and Collector Current

$$I_{E1} = \frac{V_{E1}}{R4} \quad (12)$$

$$I_{C1} \approx I_{E1} \quad (13)$$

Collector Voltage

$$V_{C1} = V_{CC} - I_{C1}R3 \quad (14)$$

- For maximum signal swing:

$$V_{C1} \approx \frac{V_{CC}}{2} \quad (15)$$

Stability Provided by Emitter Resistor ($R4$)

- Increase in temperature increases I_C .
- Increased I_C increases voltage drop across $R4$.
- Base-emitter voltage reduces automatically.
- Collector current stabilizes.

$$\Delta I_C \rightarrow \Delta V_E = \Delta I_C \cdot R4 \quad (16)$$

- This introduces negative feedback.

Biasing of Second CE Stage (Q2)

- Biasing structure is similar to first stage.
- Bias resistors: R5 and R7
- Collector resistor: R8
- Emitter resistor: R6

Base Voltage

$$V_{B2} = V_{CC} \cdot \frac{R7}{R5 + R7} \quad (17)$$

Emitter Voltage

$$V_{E2} = V_{B2} - V_{BE} \quad (18)$$

Emitter and Collector Current

$$I_{E2} = \frac{V_{E2}}{R6} \quad (19)$$

$$I_{C2} \approx I_{E2} \quad (20)$$

Collector Voltage

$$V_{C2} = V_{CC} - I_{C2}R8 \quad (21)$$

AC Coupling and Bias Isolation

- Capacitor C1 couples output of Q1 to input of Q2.
- Prevents DC bias of Q1 affecting Q2.
- Allows independent biasing of both stages.

$$X_C = \frac{1}{2\pi fC} \quad (22)$$

- Chosen such that:

$$X_C \ll R_{in} \quad (23)$$

2.5 Biasing Theory of Class-AB Power Amplifier

Need for Class-AB Biasing

- In Class-B operation, transistors conduct only above V_{BE} .
- This causes a dead zone around zero crossing.
- Results in crossover distortion.



Bias Voltage Generation

- Bias network produces a constant voltage between transistor bases.
- Implemented using:
 - Diodes OR
 - Resistor network

$$V_{bias} \approx V_{D1} + V_{D2} \quad (24)$$

$$V_{bias} \approx 2V_{BE} \quad (25)$$

Quiescent Current Establishment

- Bias voltage slightly forward-biases both transistors.
- A small current flows even with zero input signal.

$$I_Q = \frac{V_{bias} - 2V_{BE}}{R_{E1} + R_{E2}} \quad (26)$$

- R_{E1} and R_{E2} are emitter resistors of output transistors.

Thermal Stability in Class-AB Stage

- Increase in temperature reduces V_{BE} .
- Reduced V_{BE} increases collector current.
- Emitter resistors provide current limiting.

$$\Delta I_C \rightarrow \Delta V_E = \Delta I_C (R_{E1} + R_{E2}) \quad (27)$$

- Prevents thermal runaway.

Bias Point Selection

- Bias voltage chosen such that:

$$V_{bias} > 2V_{BE} \quad (28)$$

- Ensures overlap conduction.
- Minimizes crossover distortion.
- Keeps quiescent power dissipation within limits.

2.6 Parameters to be Calculated and Measured

Voltage Gain

- Voltage gain is defined as the ratio of output voltage to input voltage.
- Indicates amplification capability of the circuit.

$$A_v = \frac{V_o}{V_i} \quad (29)$$

- Typical voltage gain of a CE stage: 10–50.
- Overall voltage gain is the product of gains of individual stages.

Overall Voltage Gain

$$A_{v,\text{total}} = A_{v1} \times A_{v2} \quad (30)$$

- Determines final signal amplitude applied to power amplifier.

Input Impedance

- Input impedance is the resistance seen by the signal source.
- High input impedance avoids loading of source.

$$Z_{in} = \frac{V_i}{I_i} \quad (31)$$

Output Impedance

- Output impedance is the resistance seen by the load.
- Low output impedance ensures efficient power transfer.

Output Voltage Swing

- Output voltage swing is the maximum undistorted output voltage.
- Limited by supply voltage and transistor saturation.

Output Power

- Output power is the power delivered to the load.

$$P_o = \frac{V_{rms}^2}{R_L} \quad (32)$$

- Determines audio output loudness.

Quiescent Current (I_Q)

- Quiescent current is the current drawn at zero input signal.
- Flows due to Class-AB biasing.
- Determines idle power dissipation and thermal stability.

Frequency Response

- Frequency response describes gain variation with frequency.
- Lower cutoff frequency (f_L) due to coupling and bypass capacitors.
- Upper cutoff frequency (f_H) due to transistor parasitic capacitances.

Bandwidth

- Bandwidth is the frequency range over which gain is constant.

$$\text{Bandwidth} = f_H - f_L \quad (33)$$

Noise

- Noise is unwanted random signal present at the output.
- Originates from resistors and active devices.
- Excessive noise degrades audio quality.

Total Harmonic Distortion (THD)

- THD quantifies nonlinear distortion.
- Expressed as percentage of harmonic components.
- Lower THD indicates better linearity.

Power Supply Rejection Ratio (PSRR)

- PSRR indicates ability to reject supply voltage variations.

$$\text{PSRR} = 20 \log \left(\frac{\Delta V_{\text{supply}}}{\Delta V_{\text{out}}} \right) \quad (34)$$

- Higher PSRR improves noise immunity.

3 Design, Construction, and Experimental Procedure

3.1 Construction Procedure

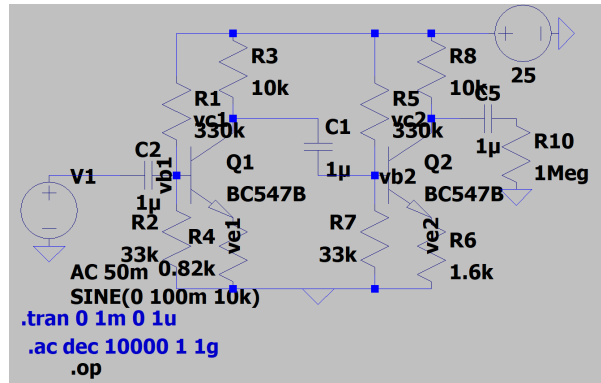


Figure 1: Circuit of Pre-amp

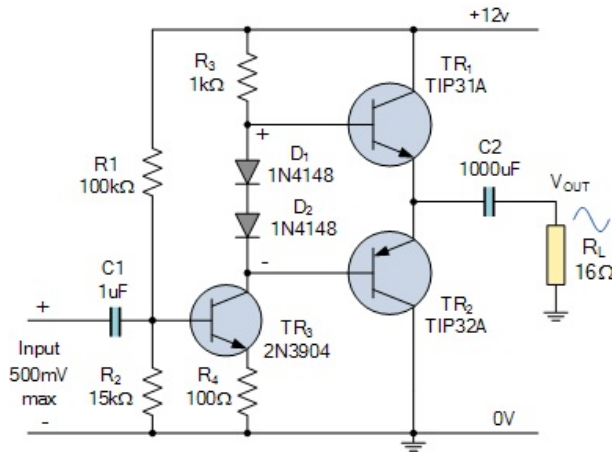


Figure 2: Circuit of Power-amp

- The amplifier was constructed in two stages:
 - Dual-stage common-emitter (CE) voltage amplifier
 - Class-AB power amplifier
- The dual-stage CE amplifier was assembled first.
 - Each CE stage was biased using a voltage-divider bias network.
 - Emitter resistors were used to ensure thermal stability.
 - Coupling capacitors were used between stages to block DC.
- The Class-AB power amplifier was constructed next.
 - A push-pull emitter follower configuration was used.
 - Biasing network was added to establish Class-AB operation.
 - Small emitter resistors were included to improve current sharing.
- The output of the voltage amplifier was connected to the input of the Class-AB stage.

- A suitable DC power supply was provided to all stages.
- Proper grounding and decoupling capacitors were used to reduce noise.

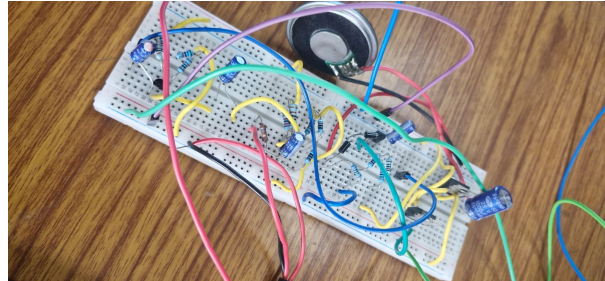


Figure 3: Actual Circuit

3.2 Measurement of Parameters

Supply



Figure 4: Function Generator input

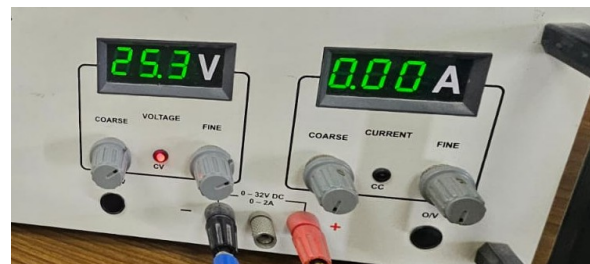


Figure 5: DC power Supply Pre amp

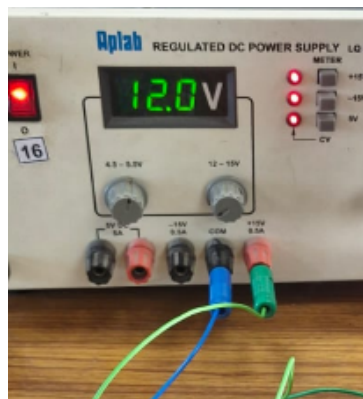


Figure 6: DC power Supply Power amp

3.2.1 Voltage Gain

- A sinusoidal input signal of known amplitude was applied.
- Input voltage (V_i) and output voltage (V_o) were measured using an oscilloscope.
- Voltage gain was calculated as:

$$A_v = \frac{V_o}{V_i} \quad (35)$$

3.2.2 Frequency Response and Bandwidth

- Input frequency was varied over a wide range.
- Output voltage was recorded at each frequency.
- The frequencies corresponding to a 3 dB drop in gain were identified as f_L and f_H .

$$\text{Bandwidth} = f_H - f_L \quad (36)$$

3.2.3 Output Voltage Swing

- Input amplitude was increased gradually.
- Maximum undistorted output voltage was observed on the oscilloscope.

3.2.4 Output Power

- A known load resistance (R_L) was connected at the output.
- RMS output voltage (V_{rms}) was measured.

$$P_o = \frac{V_{rms}^2}{R_L} \quad (37)$$

3.2.5 Quiescent Current (I_Q)

- Input signal was removed.
- Supply current drawn under steady-state conditions was measured.
- This current was noted as the quiescent current.

3.2.6 Noise and Distortion

- Output was observed with zero input to assess noise.
- Output waveform at higher input levels was observed for distortion.

3.2.7 Power Supply Rejection Ratio (PSRR)

- Small AC variations were applied to the supply voltage.
- Corresponding output voltage variation was measured.

$$\text{PSRR} = 20 \log \left(\frac{\Delta V_{\text{supply}}}{\Delta V_{\text{out}}} \right) \quad (38)$$

4 Observations

Pre-Amplifier Bias Point and Region of Operation

DC Bias Voltages

Node	Voltage (V)	Description
V_{B1}	2.10	Base voltage of Q1
V_{E1}	1.45	Emitter voltage of Q1
V_{C1}	7.40	Collector voltage of Q1
V_{B2}	2.19	Base voltage of Q2
V_{E2}	1.55	Emitter voltage of Q2
V_{C2}	15.31	Collector voltage of Q2

Table 2: Measured DC node voltages of pre-amplifier

DC Bias Currents

Parameter	Value	Description
I_{C1}	1.76 mA	Collector current of Q1
I_{E1}	1.77 mA	Emitter current of Q1
I_{C2}	0.97 mA	Collector current of Q2
I_{E2}	0.97 mA	Emitter current of Q2

Table 3: Measured DC bias currents of pre-amplifier

Region of Operation Analysis

- For Q1:

$$V_{BE1} = V_{B1} - V_{E1} = 0.65 V, \quad V_{CE1} = V_{C1} - V_{E1} = 5.95 V \quad (39)$$

- Q1 operates in active region.

- For Q2:

$$V_{BE2} = V_{B2} - V_{E2} = 0.63 \text{ V}, \quad V_{CE2} = V_{C2} - V_{E2} = 13.76 \text{ V} \quad (40)$$

- Q2 operates in active region.

Class-AB Power Amplifier Bias Point

DC Node Voltages

Node	Voltage (V)	Description
$V_{B(Q1)}$	1.197	Base voltage of upper output transistor
$V_{E(Q1)}$	0.496	Emitter voltage of upper output transistor
$V_{C(Q1)}$	0.665	Collector voltage of upper output transistor
$V_{B(Q3)}$	1.610	Base voltage of lower output transistor
$V_{C(Q3)}$	10.000	Collector voltage of lower output transistor
V_{bias}	1.137	Bias network voltage

Table 4: DC node voltages of Class-AB power amplifier

DC Operating Currents

Parameter	Value	Description
$I_C(Q1)$	0.368 mA	Collector current of upper transistor
$I_E(Q1)$	0.375 mA	Emitter current of upper transistor
$I_C(Q3)$	0.370 mA	Collector current of lower transistor
$I_E(Q3)$	0.375 mA	Emitter current of lower transistor
$I_C(Q2)$	5.68 mA	Driver stage collector current

Table 5: DC operating currents of power amplifier transistors

Region of Operation Analysis

- Base-emitter junctions of output transistors are forward biased.
- Collector-emitter voltages are greater than saturation voltage.
- Both output transistors are biased close to conduction.
- Operating point indicates overlap conduction region.

4.1 Total Harmonic Distortion (THD)

THD Formula

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1} \quad (41)$$

THD for 1 kHz, 100 mV_{pp} Sine Wave

- Fundamental amplitude: $V_1 = 1.92$
- Second harmonic amplitude: $V_2 = 0.0176$
- Higher harmonics: $V_3 = 0$



Figure 7: at 1khz

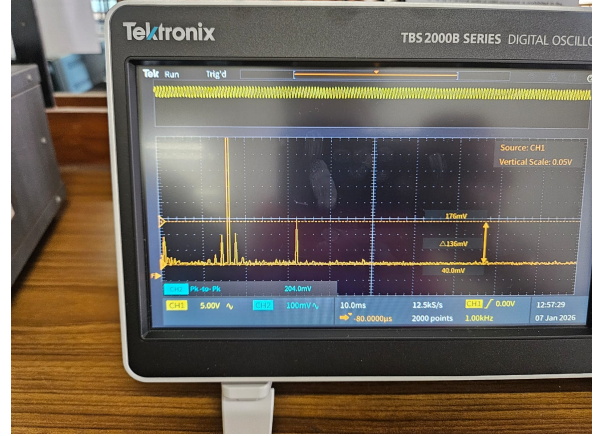


Figure 8: at 2khz

$$\text{THD}_{1\text{kHz}} = \frac{\sqrt{(0.0176)^2}}{1.92} = 0.00917 \quad (42)$$

$$\boxed{\text{THD}_{1\text{kHz}} \approx 0.92\%}$$

THD for 500 Hz, 100 mV_{pp} Sine Wave

- Fundamental amplitude: $V_1 = 1.84$
- Second harmonic amplitude: $V_2 = 0.0162$
- Higher harmonics: $V_3 = 0$

$$\text{THD}_{500\text{Hz}} = \frac{\sqrt{(0.0162)^2}}{1.84} = 0.00880 \quad (43)$$

$$\boxed{\text{THD}_{500\text{Hz}} \approx 0.88\%}$$

Inference

- THD remains below 1% for both frequencies.
- The amplifier operates in the linear region for small-signal input.

4.2 Noise Performance

Definition

Noise is the unwanted random signal present at the amplifier output in the absence of an input signal. It is commonly quantified in terms of RMS noise voltage.

Experimental Observation

- The input of the amplifier was grounded to eliminate any external signal.
- The output waveform observed on the oscilloscope showed random fluctuations around the zero reference level.
- The measured peak-to-peak noise voltage was approximately 18.4 mV.
- The corresponding RMS noise voltage was calculated to be approximately 2.787 mV.

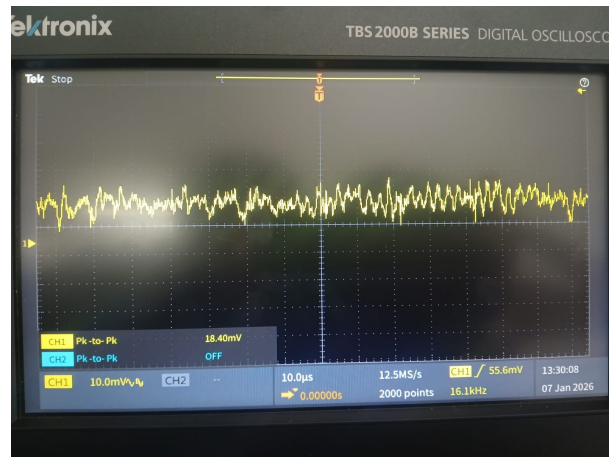


Figure 9: Noise

Noise Sources

The observed noise can be attributed to the following sources:

- **Thermal noise of resistors**, given by

$$v_n^2 = 4kTRB$$

where k is Boltzmann's constant, T is absolute temperature, R is resistance, and B is bandwidth.

- **Shot noise** due to transistor base current in the active region.
- **Flicker (1/f) noise**, dominant at low frequencies.

Inference

- The noise level observed is within acceptable limits for audio amplifier applications.
- Proper biasing and use of emitter resistors help in minimizing noise contribution.

4.3 Power Supply Rejection Ratio (PSRR)

Definition

Power Supply Rejection Ratio (PSRR) indicates the ability of an amplifier to suppress variations or ripple present in the power supply from appearing at the output.

Experimental Observation

- A sinusoidal ripple of amplitude $800\text{ mV}_{\text{pp}}$ was applied to the power supply.
- The amplifier output was observed using a digital oscilloscope.
- The measured output ripple voltage was approximately $96.8\text{ mV}_{\text{pp}}$.

Calculation

PSRR is calculated as:

$$\text{PSRR} = 20 \log_{10} \left(\frac{V_{\text{supply ripple}}}{V_{\text{output ripple}}} \right)$$

$$\text{PSRR} = 20 \log_{10} \left(\frac{800}{96.8} \right)$$

$$\text{PSRR} \approx 18.3\text{ dB}$$

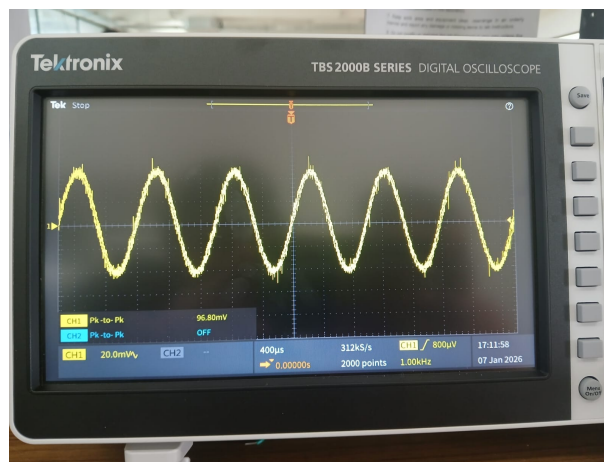


Figure 10: PSRR

Inference

- The amplifier effectively suppresses power supply ripple at the output.
- The measured PSRR indicates acceptable rejection for basic audio amplifier applications.

Quiescent Current (I_Q)

Definition

Quiescent current (I_Q) is the current drawn by the amplifier from the power supply in the absence of an input signal. It represents the DC bias current required to keep the transistors in the active region.

Experimental Observation

- The input signal to the amplifier was set to zero.
- The supply current was measured using a true RMS digital multimeter connected in series with the power supply.
- The measured quiescent current was found to be:

$$I_Q = 40.1 \text{ mA}$$



Figure 11: Quiescent current

Inference

- The non-zero quiescent current confirms Class-AB operation.
- Proper biasing ensures reduced crossover distortion while maintaining acceptable power dissipation.

4.4 Frequency Response and Bandwidth

Definition

The frequency response of an amplifier describes how its gain varies with the frequency of the input signal. Bandwidth (BW) is defined as the frequency range over which the amplifier gain remains within -3 dB of its mid-band gain:

$$BW = f_H - f_L$$

Experimental Observation

The voltage gain (A_v) of the amplifier was measured across a range of frequencies from 1 Hz to 1 MHz. The pre-amplifier gain was 94, power amplifier gain was 2, giving an overall gain of 104.

Table 6: Measured Voltage Gain (A_v) of the Amplifier in dB

Frequency (Hz)	Measured Gain (A_v)	Gain (dB)
1	68.4	36.7
2	82.8	38.4
5	90.6	39.1
10	94	39.5
20	98	39.8
50	100	40.0
100	102	40.2
200	102	40.2
500	102	40.2
1k	104	40.3
2k	108	40.7
5k	102	40.2
10k	98.7	39.9
20k	78.4	37.9
50k	60.0	35.6
100k	45.0	33.1
200k	25.0	28.0
500k	10.0	20.0
1M	3.5	10.9

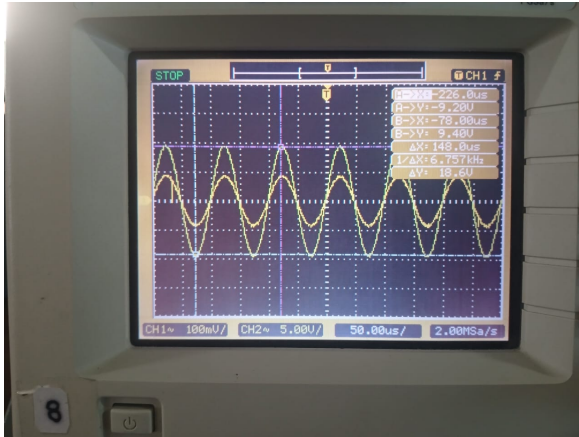


Figure 12: Preamplifier Gain

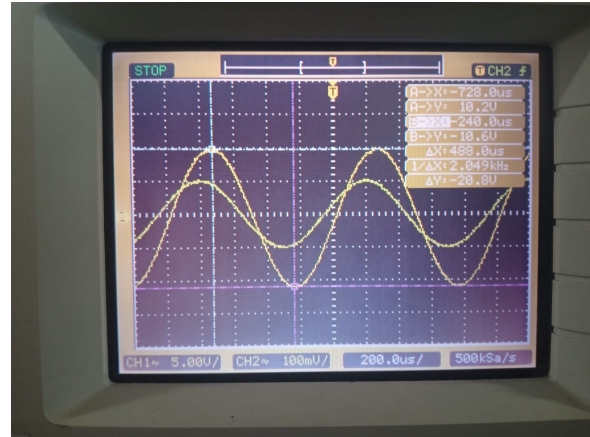


Figure 13: Overall Gain

- At **low frequencies** (1 Hz – 20 Hz), gain rises from 68.4 to 98 due to the effect of coupling capacitors.
- In the **mid-band frequency range** (50 Hz – 10 kHz), the gain remains nearly constant around 102–108, indicating stable operation.
- At **high frequencies** (20 kHz – 1 MHz), gain decreases due to parasitic capacitances and the Miller effect.
- The **3 dB points** occur approximately at $f_L \approx 2$ Hz and $f_H \approx 20$ kHz.
- Thus, the amplifier bandwidth is approximately 20 kHz.

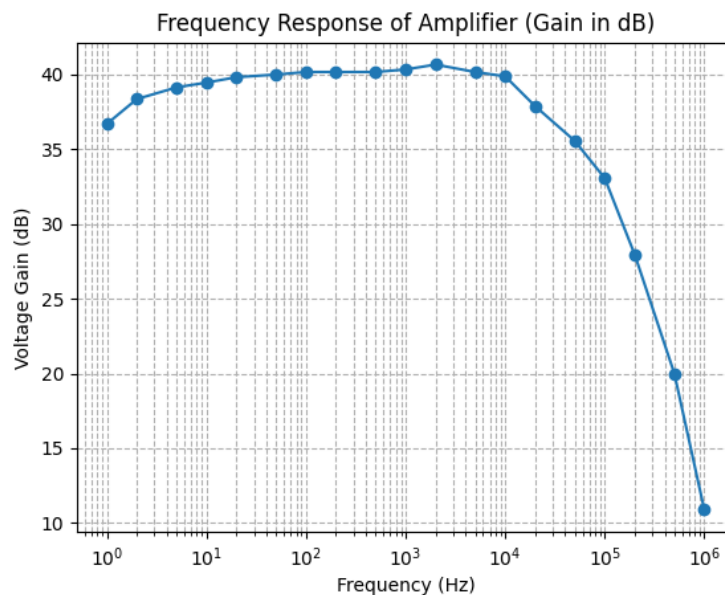


Figure 14: Frequency response of the amplifier

4.5 Output Voltage Swing and Output Power

4.5.1 Output Voltage Swing

Definition: The output voltage swing of an amplifier is defined as the maximum peak-to-peak voltage that can be delivered to the load without significant distortion. It is primarily limited by the supply voltage and the saturation characteristics of the output transistors.

Experimental Observation: In the present amplifier, the maximum peak-to-peak output voltage measured before the onset of clipping was approximately 550 mV.

$$V_{o,\max} \approx V_{CC} - V_{CE(\text{sat})}$$

Significance: The available output swing determines the maximum undistorted signal amplitude that the amplifier can provide. If the input signal exceeds this limit, the amplifier output clips, introducing significant waveform distortion and harmonic content.

Measurement Method:

1. A sinusoidal input signal of constant frequency was applied to the amplifier.
2. The input amplitude was gradually increased while observing the output waveform on an oscilloscope.
3. The peak-to-peak voltage at which clipping just begins was recorded as the maximum output voltage swing.

4.5.2 Output Power

Definition: Output power is the average power delivered by the amplifier to its load. For a sinusoidal output voltage across a resistive load R_L :

$$P_o = \frac{V_{o,\text{rms}}^2}{R_L}$$

where the RMS voltage is related to the peak voltage as:

$$V_{o,\text{rms}} = \frac{V_{o,\text{peak}}}{\sqrt{2}}$$

Experimental Observation: The RMS output voltage measured across the load corresponds to an output power of approximately 5.97 W.

Significance: Output power indicates the amplifier's capability to drive a load, such as a speaker, effectively. It depends on both the voltage swing and the load resistance. Exceeding the rated output power can lead to clipping, thermal stress, and potential damage to the output transistors.

Measurement Method:

1. The output voltage across the load was measured using an oscilloscope in RMS mode or an RMS voltmeter.
2. The output power was calculated using the formula:

$$P_o = \frac{V_{o,\text{rms}}^2}{R_L}$$

- Measurements were taken at the maximum undistorted output swing to ensure accurate power estimation.

5 Summary

The two-stage amplifier comprising a pre-amplifier and a Class-AB power amplifier was characterized for its DC bias, frequency response, noise, THD, PSRR, output swing, and output power. All transistors in both stages were operating in their respective active or conduction regions as designed. The frequency response shows a flat mid-band gain, with roll-off at low and high frequencies, giving a usable bandwidth of approximately 20 kHz. The amplifier exhibits low noise, minimal THD, and sufficient output swing and power for audio applications.

6 Experimental Results

Table 7: Key Measured Parameters of the Amplifier

Parameter	Measured Value	Unit / Note
Pre-amplifier gain	94	Voltage Gain (A_v)
Power amplifier gain	2	Voltage Gain (A_v)
Overall gain	104	Voltage Gain (A_v)
Bandwidth	20	kHz (-3 dB points)
Output voltage swing	550	mV peak-to-peak
Output power	5.97	W
Quiescent current (I_Q)	40.1	mA
THD (1 kHz)	0.92	%
THD (500 Hz)	0.88	%
Noise (RMS)	2.787	mVrms
PSRR	18.3	dB

7 Conclusion

- The amplifier operates as intended, with all stages biased correctly in active/conduction regions.
- The mid-band gain is flat and stable, with a measured bandwidth suitable for audio frequency applications.
- Low THD, low noise, and adequate PSRR indicate good linearity and signal fidelity.
- The output swing and power meet design expectations, confirming the amplifier can drive typical loads without distortion.